1.The Funds and Backtest Design

1.1The Mechanics of Walk-Forward Backtesting and Look-Ahead Bias

Using only the data available at that exact moment in time, a walk-forward out-of-sample backtest is essentially a rigorous simulation of how a trading strategy would have performed in real life. Think about managing a portfolio on a monthly basis. For example, on January 1st, you are limited to make decisions based on past data that is available thru December 31st. This procedure is carried out in NoBearTrader's particular implementation by employing a rolling estimating window along with monthly rebalancing. The optimiser determines previous asset returns, volatilities, and pair-wise correlations based on a trailing period of 252 trading days for traditional assets, or 365 days for cryptocurrencies traded around the clock. The model bases its decision on target weights on the trailing period alone, holds these weights fixed and compares them to the actual market return in the following month, advancing by a month before starting the optimisation process again.

However, the output of such backtesting will become entirely fabricated if this strict temporal boundary is crossed because of an inherent flaw in the structure known as "Look-Ahead Bias." To put things in perspective, when a mathematical model calculates portfolio weights for January and uses price information from February, it is actually looking ahead into the future. Moreover, a look-ahead-biased model will magically sell or underweight an asset on January 1st if it experiences a significant crash in the middle of January, since the model has already anticipated the crash.A look-ahead-biased model will suddenly sell or underweight an asset on January 1st if it experiences a significant crash in mid-January because it anticipates the crash.On paper, this may seems like it creates a smooth, risk-free profit curve that appears to be a surefire way to make money, but when a retail investor uses the approach in the real world, it will fail instantly. Hence, in order to make sure the backtest accurately reflects out-of-sample risk and viability is to evaluate portfolio strategies under strict walk-forward constraints because real market participants do not have access to tomorrow's newspaper.

1.2 Optimisation Objectives and Portfolio Frameworks

NoBearTrader uses four separate optimisation goals to build the quantitative funds, each of them represent a fundamentally different method of portfolio development:

- **Equal-Weight (Naive Allocation):** This is considered as the most straightforward method, in which an investor distributes the same amount of money among all accessible assets ($w_i = 1/N$). It relies on the empirical principle that naive diversification offers an exceptionally strong defence against estimating error and market uncertainty as well as ignoring past return and volatility data.

- **Minimum-Variance:** By determining asset weights that optimise diversification as well as counteract individual price fluctuations, this approach aims to create the portfolio with the lowest possible volatility for investors. It naturally favours stable, defensive assets with low pairwise correlations since it puts capital preservation and smooth equity curves priority over pursuing maximum profits.
- **Maximum-Sharpe (Tangency Portfolio):** This approach is able to maximise the projected risk-adjusted return ratio (excess return per unit of total portfolio volatility) as opposed to concentrating only on risk minimisation. Although it actively seeks risk efficiency, it frequently makes highly concentrated, aggressive bets on recent high-performing assets due to its substantial dependence on estimating trailing average returns.
- **Risk Parity:** Rather than prioritising on equalising the monetary amounts that have been invested in each asset, for example Equal-Weight, this approach is focusing on equalising the volatility contribution of each asset to the portfolio's overall variance. Besides, it guarantees that no single high-volatility asset dominates the fund's overall risk profile, which means safer assets receive bigger capital allocations while highly volatile assets receive smaller, risk-controlled weights.

1.3 Comparative Asset Universes

By building these portfolios across three different asset universes such as equity-only, crypto-only as well as combined. It allows investors to isolate the raw behaviour of each assets class before they interact within a single portfolio:

- **Equity-Only Universe:** Functions as the conventional market baseline, comprising well-known, liquid single-stock equities from a variety of macroeconomic markets.
- **Crypto-Only Universe:** This universe captures the dynamic continuous, high-volatility, round-the-clock trading of decentralised digital assets.
- **Combined Universe (60 Assets):** A single multi-asset framework that is created by combining conventional stocks and cryptocurrencies.

In addition, an investor would have no empirical way to find out whether combining traditional and decentralised assets can actually improve risk-adjusted performance by implementing cross-asset diversification, or if one asset class simply dilutes and drags down the performance of the other without the equity-only and crypto-only baselines serving as strict control groups.

1.4 Rebalancing Frequency and Risk-Free Rate Baseline

In order to achieve a balance between the potential risk of high frictional churn and the capture of short-term market movements, a monthly rebalancing was chosen as a typical, useful baseline. Furthermore, by adjusting portfolio weights on a monthly basis able to prevent the high transaction costs of weekly or daily re-optimisations while giving quantitative signals enough time to materialise.

On top of that, it was a deliberate, standard modelling decision to set the risk-free rate to zero (rf=0) in all Sharpe ratio computations. In this way, the methodology allows us to effectively isolate the purely relative architectural efficiency of the optimisation processes as well as streamline cross-sectional comparisons among all 12 funds. Before proceeding any further in our process of conducting transaction cost stress tests and comparing performance against external rate benchmarks, the methodology ensures that any performance increases are due to pure allocation skill.

<u>1.5 Solver Stability and Ridge Regularization</u>

The use of ridge regularisation method (_ridge_cov()) was incorporated into the covariance estimation process to mitigate the operational risk that may exist in cases where the numerical optimisers may face challenges when dealing with the highly correlated data set for the daily returns. Simply put, ridge regularisation applies small penalties on the diagonal of the sample covariance matrix ($\Sigma reg = \Sigma + \lambda I$).

In addition to this, this structural change ensures that the covariance matrix is going to be always positive-definite and invertible rather than turning out to be singular or ill-conditioned. In case of the 60-assets Combined universe dataset, wherein the relatively small estimation period of 252 days leads to a high-dimensional, thin data-to-parameter ratio problem, ridge stabilization has become essential. Conventional solvers of quadratic programs will not work without ridge stabilisation in the presence of even minor changes in the trailing returns of the assets

2. Out-Of-Sample Results and Fund Fact Sheets

<u>2.1 Headline Performance Across Universes</u>

Cross-asset diversification has resulted in several interesting outliers during the 2021-2023 out-of-sample phase depending on the optimisation approach. In particular, the simplest Equity Equal-Weight portfolio has shown itself to be extremely solid as a baseline in conventional markets (Sharpe 0.782), which means that simple allocation works very well out-of-sample. Besides, although approaches such as Crypto Risk Parity seem not to work as well (Sharpe 0.475), pure cryptocurrency funds have been doing great in this regime, with Crypto Minimum-Variance leading the pack (Sharpe 0.912).

The combination of such asset classes would lead to great performance divergence, which is the most striking pattern. In fact, some of the models have shown great stress in the structure due to the pushing of assets with different volatility through the conventional optimizers: A negative risk-adjusted return (Sharpe -0.271) is the consequence of Combined Minimum-Variance's extreme troubles. However, on the contrary, the Combined Maximum-Sharpe fund obtained a Sharpe of 1.092 which is the highest among all 12 funds as it efficiently exploits the diverse volatility. Additionally, Combined Risk Parity has managed to pass through the mixed universe with a great Sharpe of 0.861. This proves that although combining traditional and decentralised assets can result in outstanding results, the mathematical objective of the fund plays an important role. This difference has been proven by the growth-of-$1 chart in Figure 1, where the great fluctuation of the crypto-only funds is seen to exceed those of equity and combined funds. Meanwhile, the best-to-worst dispersion becomes evident straight away due to the Sharpe ratio (Figure 4).

2.2 Diagnostic: Combined Minimum-Variance Instability

The model has revealed an exceptionally high turnover while calculating the Combined Minimum-Variance portfolio, achieving a median value of about 106% monthly, which appears to be a technical bug in programming. Yet, because of the low proportion of the number of days in the moving window to the number of parameters of the algorithm that seeks to rebalance the 60 securities in a 252-day window, it leads to the mathematical instability of the covariance matrix optimisation and causes this turbulence. Nonetheless, this instability allows the optimiser to shift its money across economically reasonable defensive stocks like Walmart (WMT), Coca-Cola (KO), Duke Energy (DUK), and Merck (MRK). Eventually, the model is forced to get stuck into an infinite cycle of costly defensive rebalancing because of its extremely sensitivity to any variations in the historical volatility. In addition, the consequence of this instability is illustrated in Figure 2 below by demonstrating several sharp declines in the drawdown of the fund instead of one decline only.

2.3 Diagnostic: Maximum-Sharpe Concentration Risks

On the contrary, with a Sharpe ratio of just -0.241, the Crypto Maximum-Sharpe portfolio was the worst in terms of risk-adjusted returns across all cryptocurrency-focused funds. An excessive concentration on single assets is the key factor behind such a poor performance. The fund model tends to invest 100 percent of its funds in a single asset in most diversified months and never goes lower than a maximum allocation of 36% even at its most diversified month. Besides, by utilising the algorithm on an extremely small sample size of only ten assets with a very noisy 365-day lookback period, it makes the optimizer generate a fragile, concentrated portfolio that quickly becomes inefficient out-of-sample due to the fact that it seeks maximum return-to-risk ratio.

An extreme concentration exists in both the underperforming crypto fund and the top-performing combined maximum-sharpe portfolio, which has had a 48.97% investment in one particular stock (GE_EQ) recently. This demonstrates that extreme fragility is not a problem with poor selection of assets but a result of the structural flaws of maximum-sharpe optimisation in itself. It demonstrates to an investor that having a high Sharpe ratio may not necessarily mean that the investor has a good handle on risk-adjusted returns but merely a one-off lucky bet with a very high level of idiosyncratic risk. Ultimately, evaluating the performance of a fund through the Sharpe ratio is dangerous since the ratio is purely backward looking as well as it ignores the idiosyncratic risk present.

<u>2.4 Fact Sheet Profile: The Combined Maximum-Sharpe Fund</u>

The investor who has gained access to the interactive fact sheet for the Combined Maximum-Sharpe fund will see some excellent headline performance figures where a one-dollar initial investment made in January 2021 grew to a remarkable of $2.05 by the year-end 2023. This leads to an exceptionally efficient Sharpe ratio of 1.09 and a very decent annualised return rate of +27.2%. Nevertheless, a different risk tale can be uncovered once the investor examines the inner workings of the portfolio of this fund, namely, the devastating maximum loss of -23.1%.

Furthermore, another proof of logic both of its outstanding performance and its vulnerability lies in its breakdown of holdings. It is important to note that its latest rebalance showed the portfolio's extremely concentrated position in one equity stock, which is General Electric (GE_EQ) holding about half of the capital amounting to 48.97%, plus its aggressive and excessive bets on NVIDIA (NVDA_EQ) and Bitcoin (BTC-USD_CR), which are equal to 18.32% and 11.27% respectively. It is obvious from the breakdown of holdings of the fund represented in Figure 3, where the bar of GE_EQ dominates the rest of the portfolio. Taken together, these numbers give a potential investor a complete understanding. Moreover, through aggressive betting on a couple of high momentum assets in both traditional and decentralised financial markets, it generated a performance superior to that of the market. However, it needs an investor who is willing to suffer substantial drawdowns.

<u>2.5 The Consistency of Risk Parity</u>

The analysis of the Risk Parity strategy in all three asset pools shows an unmistakable tendency towards consistency, which surpasses even the optimal and worst performing funds. While the Minimum-Variance strategy showed negative out-of-sample performance due to extreme turnovers (negative Sharpe ratios in two out of three universes), the Maximum-Sharpe ratio strategy demonstrated an erratic performance ranging from an extremely good 1.092 in the combined pool to an extremely poor -0.241 in Crypto universe. The Risk Parity strategy demonstrated Sharpe ratios of 0.475, 0.696, and 0.861 in crypto, stocks, and combined universes respectively.

This constant performance demonstrates that when connecting different types of asset classes, using the “equal risk contribution” approach is actually much better. Other than the fact that the pure return chaser optimiser approach creates very fragile bets which have very little value, the Risk Parity approach makes it impossible for hyper-volatile assets like Bitcoin to dominate the portfolio’s returns by ensuring that each individual asset contributes the same amount of volatility to the portfolio as a whole. This can be noted from Figure 5 where the concentrated bets of the Maximum-Sharpe portfolio are compared to those of the Combined Risk Parity approach.

3. The Sentiment Index

3.1 The Need for a Domain-Specific Lexicon

In terms of structure, it is flawed as it depends entirely on using an off-the-shelf and generic sentiment analysis tool like basic VADER for analysing finance-related text because the plain VADER was trained on social media language, and thus it inherently fails in the presence of the unique lexicon of financial news.(Hutto & Gilbert 2015) Furthermore, there are many accounting terms such as "liability" and "tax" which have been wrongly interpreted as negative by generic emotion dictionaries.

This misalignment has become very evident when examining the structural reporting terminology. For instance, a general sentiment tool often grants the word "buy" a significant positive valence. However, referring back to Part A's corpus analysis, "buy" is the most common phrase in the dataset, serving mostly as a format-driven reporting tag (e.g., "Analyst assigns Buy rating") rather than conveying true market excitement. Therefore, in order to prevent this structural noise from artificially boosting the index, there are terms like "buy" and "sell" that have been manually removed from the custom finance-extended lexicon. Based on empirical evidence, the standalone VADER compound score of the term "buy" was successfully lowered from 0.2023 to 0.0 by neutralising it. This was achieved by eliminating its entry completely, so the model reverts to treating it as an unscored, neutral word. In addition, instead of stumbling over the grammatical artefacts of financial journalism, this intentional improvement is able to ensure that the final emotion index represents true psychological market fluctuations.

3.2 Constructing the Sector Index

In order to develop the sector-level sentiment index, the raw headline scores are first averaged per ticker for a particular day. In addition, the daily sector index is created using equal weighting across all tickers within the sector. On top of that, by taking into account the neutrality of no-news ticker-days (which means setting their value to 0) is an important part of the methodology used in this aggregating process. Furthermore, the volatility of the index will be falsely high on a day-to-day basis if no-news ticker-days are simply taken out of the data set because it allows for a single headline about a small ticker to control the whole average of the sector in the case where there is nothing else going on. The program properly accounts for media silence as a lack of sentiment by assigning neutral values to the missing days.

3.3 Visualizing the Sentiment Landscape

The heat map representation of the sector sentiment is one which is largely composed of a positive or 'green' area, which proves the conclusion mentioned in part A , in which only 7.7% of the financial corpus contained clearly sentimental terms. In other cases, when there have been drops in the market, the index hardly shows any deep red or negative sentiment as financial reporting tends to use structural, objective words in describing events rather than emotional language. Nevertheless, the index is capable of effectively distinguishing the actual changes in the economy despite the positive biasness in the structure of the index. This heat map is presented below in figure 6, where an essentially neutral pale green background is contrasted with a dark green ring of increased emotion around the Industrials and Real Estate sectors in 2022-2023. This prolonged peak demonstrates that the custom lexicon can successfully map real-world economic narratives by precisely reflecting the post-pandemic increase in logistics, supply chains, and industrial warehousing capacity.

4. Extensions and Innovations

4.1 Systematic Sentiment Fusion and Results

The pipeline uses a systematic sentiment tilt that computes a cross-sectional z-score over a rolling lookback window, actively shifting baseline portfolio weights toward assets experiencing abnormally positive news so that it is able to successfully incorporate unstructured textual data into the quantitative models.(UNSW Business School 2026) In order to rigorously avoid any look-ahead bias, our fusion mechanism is fully lag-safe, guaranteeing that sentiment signals only affect following trading days. Moreover, by applying this innovation, it has resulted in out-of-sample risk-adjusted improvements across all three tested baselines. First, the Combined Maximum-Sharpe has experienced a marginal improvement from 1.092 to 1.097, while the Equity Maximum-Sharpe fund increased its Sharpe ratio from 0.607 to 0.651. On top of that, the most significant transformation occurred within the previously unstable Combined

Minimum-Variance fund, where the sentiment tilt completely reversed a negative baseline Sharpe of -0.271 into a robust 0.578. This demonstrates that alternative data can successfully anchor and rescue structurally brittle optimisations.In Figure 7, it visualises this striking outcome by showing the base-versus-fused growth trajectory for Combined Minimum-Variance; throughout the out-of-sample window, the fused line consistently and clearly pulls ahead of the base line.

4.2 Sensitivity Grid and Parameter Stability

The fusion method was exposed to a full parameter sensitivity grid over nine different combinations of tilt strength and rolling lookback windows in order to mitigate the potential that sentiment adjustment simply overfits historical data—a failure scenario clearly shown in the course baseline. Most importantly, before any performance outcomes were assessed, the model's default parameters(a tilt strength of 0.25 and a 5-day lookback) were locked in as untuned defaults.

Tilt Strength	Lookback (Days)	Sharpe Ratio
0.10	3	0.613
0.10	5	0.607
0.10	10	0.594
0.25	3	0.592
0.25	5(default)	0.578
0.25	10	0.545
0.50	3	0.568
0.50	5	0.549
0.50	10	0.467

According to the grid above, it shows that every combination of parameters has resulted in a favourable performance gain, with the Sharpe ratio presenting a smooth, monotonic decrease as both tilt strength and lookback length increased. This predicted surface shows a smooth performance fall as news signals age or are overweighted. Moreover, since this surface automatically filters out weak and noise-induced peaks, this proves that the fusion picks up an actual economic news signal rather than just a lucky combination of parameters.

4.3 Robustness under Transaction Costs

In two out of three portfolios that have been tested, the superior performance of the fusion continued through a stringent 50 basis point transaction cost sensitivity analysis. Although the friction successfully stripped the highly marginal gross advantage of the Combined Maximum Sharpe fund, the friction managed to preserve the huge advantages in the Combined Minimum Variance fund, where the sentiment tilt inadvertently lowered turnover.

4.4 The Lexicon Advantage

Both the default model of the course and the rolling z-score tilt have a same structure, but the level of the purity of the information contained in the sentiment is the key factor that distinguishes these two. (UNSW Business School 2026) Before applying the tilts mathematically, the system filters out any false positives, which results from the replacement of the default VADER with the financial lexicon of our own, which excludes such non-informative terms as “buy” and “sell”. It is this high quality information that makes the trading strategy successful.

5. The App and Investor Journey

5.1 The NoBearTrader Product Experience

As soon as an investor gets onto NoBearTrader website, they will start off with the Funds & fact sheets page, where one can see how all the twelve quantitative methods are compared before going deeper into analysis of each separate fund and its growth, drawdown, and holdings in order to understand its unique risk profile. Further going to the Build Your Allocation tab, an investor is able to mix different funds by means of the interactive sliders and even to take control and create a custom out-of-sample portfolio, matching the investor’s unique risk tolerance level. In addition to that, investors can visit to the Sentiment tab in order to look at the news-sentiment index for each sector, which forms the foundation for quantitative models, so that one could understand what is the proprietary edge in these profits. Lastly, in the Data inventory tab, there is an entire transparency, where one is able to see the raw price and headlines data, which form the basis for the entire ecosystem of this product.

5.2 Real-World Testing and Quality Assurance

While performing testing of the application before its deployment, there were specific portfolios that showed a growth number of “$nan” along with inaccurate performance due to a major calendar mismatch error. The error in question happened due to two interconnected issues stemming from the same problem with the calendar. Those issues include the hard-coded factor of 252 days for annualization not considering that some portfolios have high crypto asset allocation with a 365 days calendar and dates when crypto assets trade but no equity data is available, resulting in “NaN” being injected into the return calculation. Besides solving the issue, by restructuring the code in order to make calendar adjustments and apply different annualization factors for different universes proved that quality control is crucial in building a production-grade solution with AI help.

5.3 UI/UX Refinement and Visual Integrity

Systematic UI/UX testing was necessary to guarantee that the program was not only mathematically functional but also truly user-friendly and reliable for a retail investor, going beyond fundamental algorithmic stability. A clearly specified visual hierarchy was necessary to restore contrast and clarity after early deployments showed that a lacking theme configuration resulted in washed-out text labels and unreadable black native charts. A structural redesign was necessary to divide the visualisations into two separate, comprehensible panels since trying to plot all twelve quantitative funds on a single growth chart had resulted in significant visual compression that made the performance data unreadable. Lastly, the daily noise of the raw sector-sentiment score made it difficult to identify underlying macroeconomic changes. To navigate these issues, by incorporating an interactive smoothing slider, the interface enabled users to filter out high-frequency journalistic noise, highlighting that effective deployment necessitates bridging the gap between practical, user-centric design and raw numeric data.

6. Critical Reflection with Three Concrete Recommendations

6.1 Project Reflection

The most important lesson that I have learned from developing NoBearTrader is that conventional mathematical optimisation frequently produces a risky appearance of confidence as well as conceals serious structural weakness underneath alluring top-line measures. It can be a behavioural trap to define success by using only a backward-looking metric like the Sharpe ratio, which presents an undiversified, highly concentrated wager,for example a nearly 50% allocation to a single stock as a better risk-adjusted result. Ultimately, algorithms lack economic sensibility and only exploit the mathematical limitations placed upon them, underscoring the ongoing necessity of strict human supervision and moral diagnostic testing when creating financial products. This procedure has clearly shown that quantitative models cannot be relied upon automatically. Hence, in order to distinguish genuine skill from statistical chance, a comprehensive, qualitative investigation of a portfolio's actual out-of-sample behaviour is required.

6.2 Specific Recommendations for Future Development

In light of the diagnostic outcomes as well as the structural weaknesses revealed during this research, the next versions of the NoBearTrader trading pipeline should feature the following three distinctive upgrades:

1. **Set strict position-size limits in the Maximum-Sharpe optimisation constraints:** Enforcing a strict maximum weight threshold, for example 15% per asset, the mechanism will automatically preclude any dangerous and highly undiversified exposure to the noisy historical price series owing to extreme concentration risks discovered, such as the 48.97% weighting of GE_EQ and 100% crypto single-position bets.
2. **Apply a turnover-aware threshold rule for rebalancing:** In order to stabilise the highly volatile median monthly turnover rate for the Combined model of approximately 106%, it is necessary to design the "no-trade zone" procedure. Furthermore, it is required to eliminate any costly micro-adjustments since it will allow ensuring the rebalancing is performed if the new target weights calculated by the optimiser significantly deviate from the current ones.
3. **Integrate a live transaction-cost penalty directly into the optimizer's objective function:** If the execution cost is included mathematically in the optimiser itself, the models will be driven automatically to optimize their returns net of fees, thereby avoiding the over-trading behavior that currently destroys their performance theory, rather than imposing a simple 50 basis point friction as a stress test.

6.3 Conclusion

To sum up, more is required from combining traditional and decentralized assets in an unregulated mathematical optimizer in order to fulfill the basic premise behind NoBearTrader’s “no bear” portfolio. It can only happen when these complex algorithms are bound by economic common sense and backed by highly accurate alternative data in case of failure of historical numbers.

Appendix

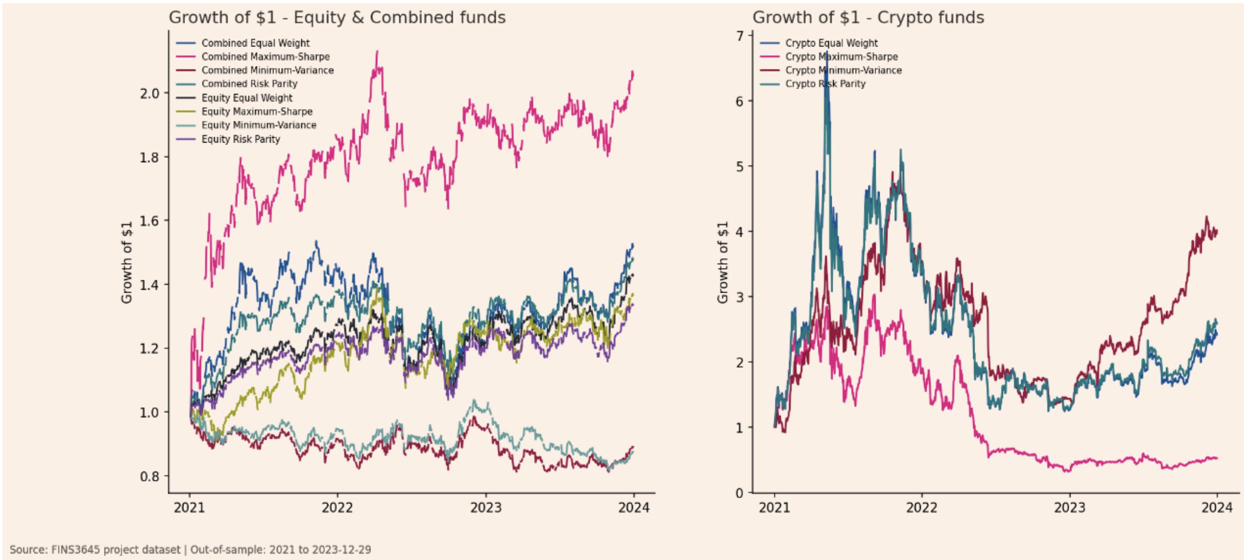

Figure 1. Growth_of_dollar_all_funds

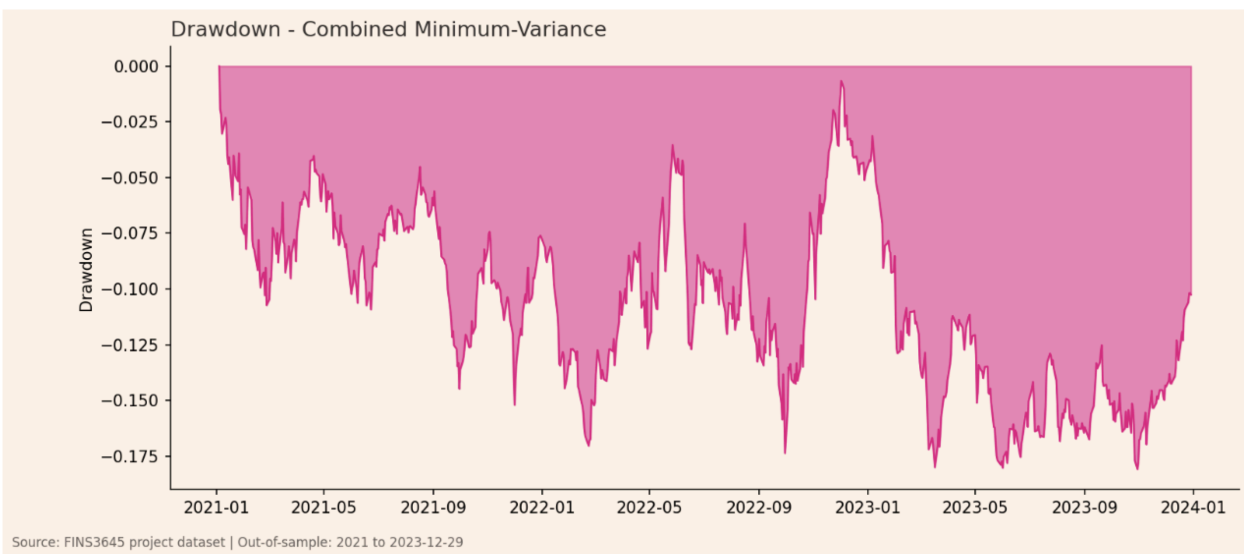

Figure 2. Drawdown_combined_min_variance

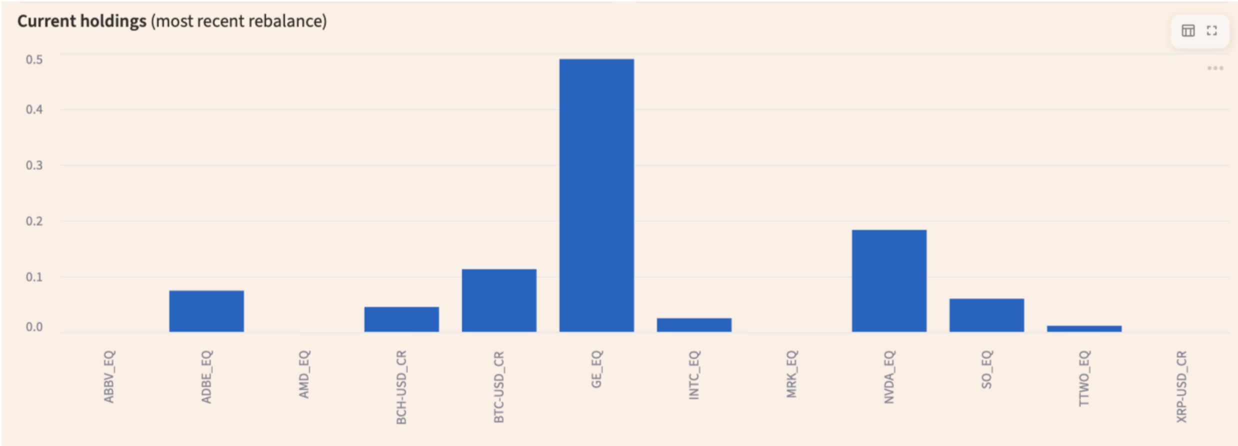

Figure 3. Current holdings, Combined Maximum-Sharpe fund fact sheet (most recent rebalance). Source: NoBearTrader Streamlit app.

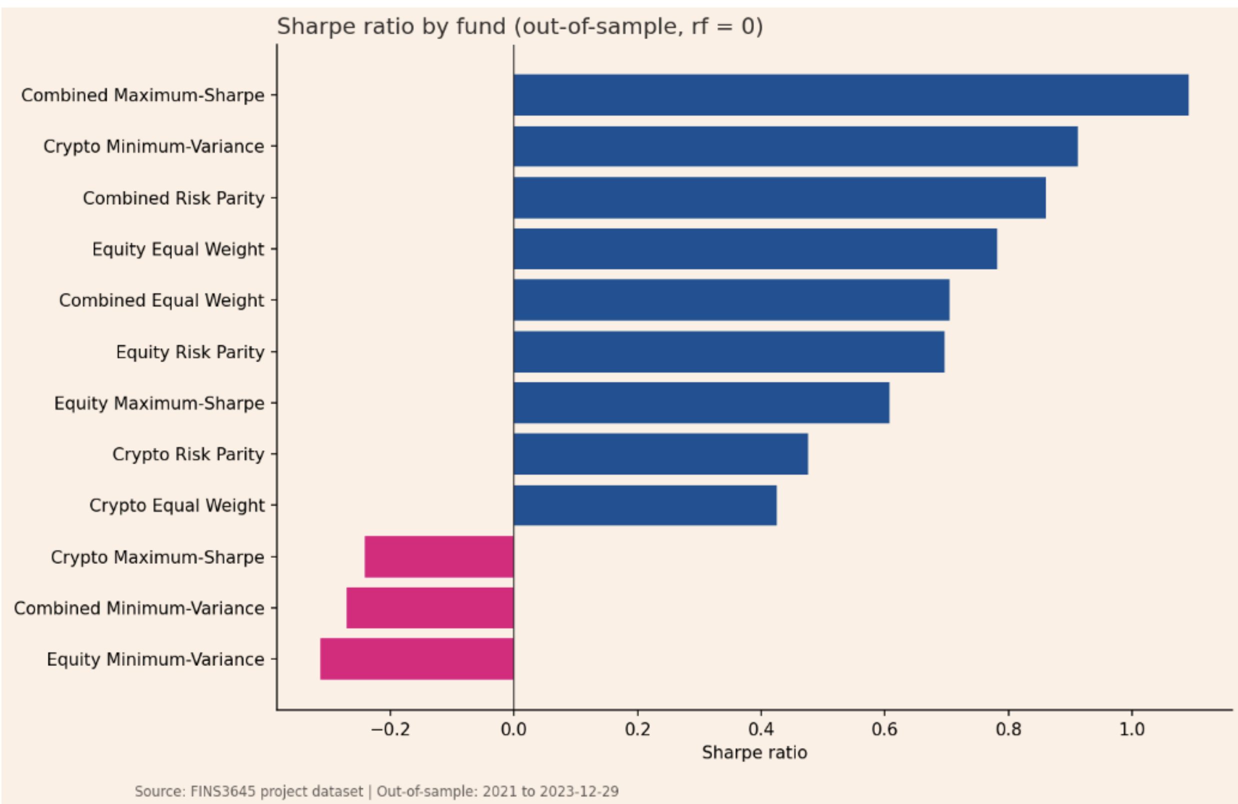

Figure 4. Sharpe_by_fund

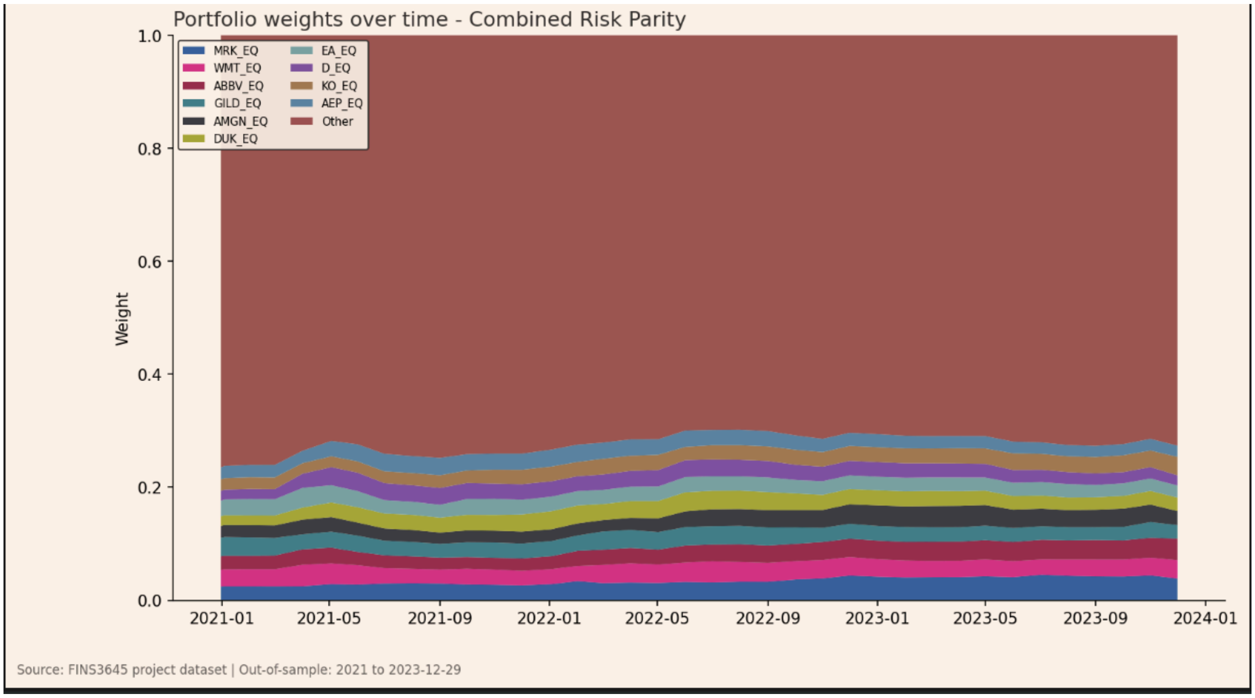

Figure 5. Weights_combined_risk_parity

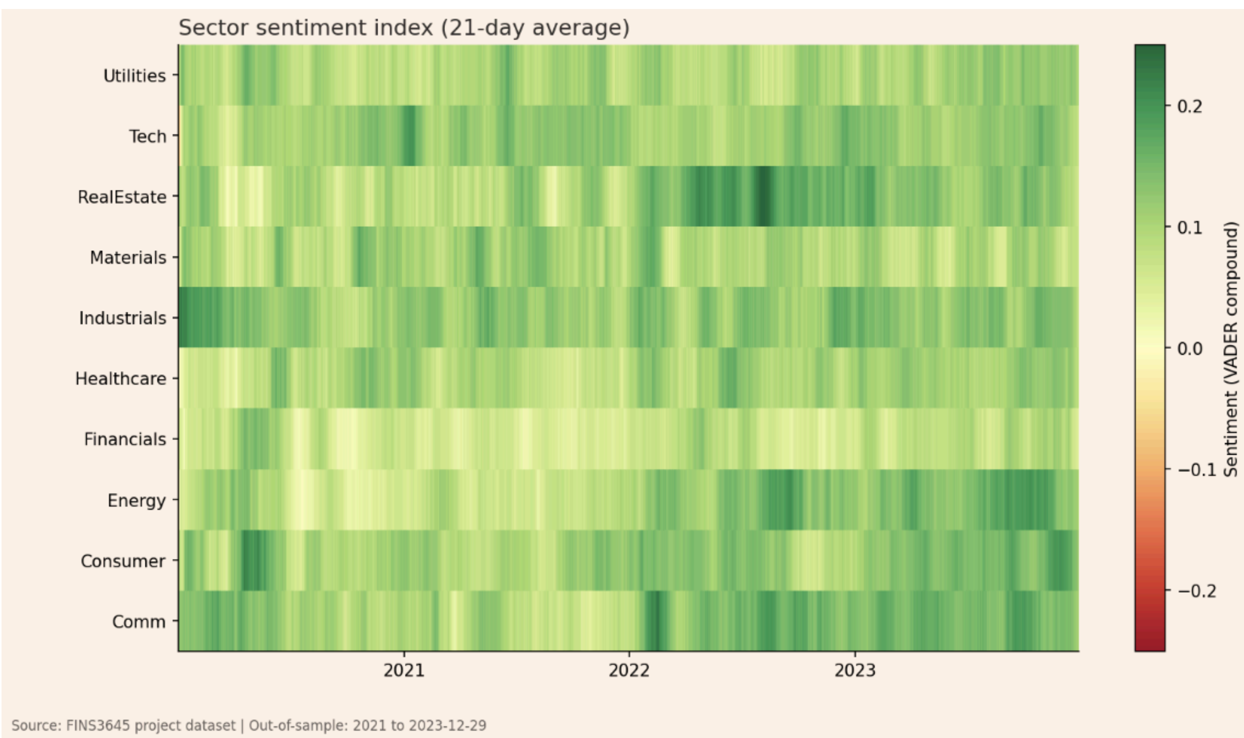

Figure 6. Sector_sentiment_heatmap

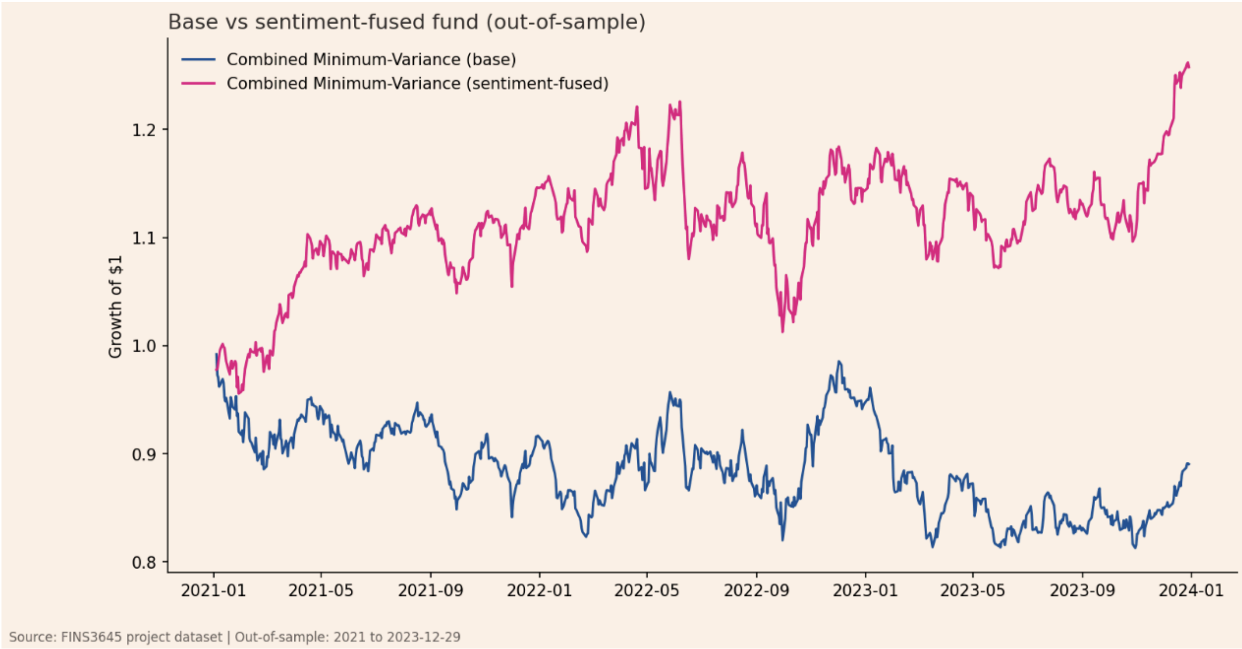

Figure 7. Base vs. sentiment-fused growth of $1, Combined Minimum-Variance (out-of-sample)

Deployment Links

Live app: https://z5608063projectb-9gzquappdkyiqdvjqqwcc7y.streamlit.app

Public repository: https://github.com/WeiKeatGan/z5608063_projectB

Reference Lists

FINS3645 Course Team 2026, *FinTech Project 2026 dataset: equity_prices, crypto_prices, news_headlines*, data set, UNSW Sydney, School of Banking and Finance.

Hutto, CJ & Gilbert, E 2015, *(PDF) vader: A parsimonious rule-based model for sentiment analysis of Social media text*, *ResearchGate*, January, viewed 24 July 2026, <https://www.researchgate.net/publication/275828927_VADER_A_Parsimonious_Rule-based_Model_for_Sentiment_Analysis_of_Social_Media_Text>.

UNSW Business School 2026, *FINS3645 Week 10 lecture: Agentic Coding and Revision*, lecture notes, School of Banking and Finance, UNSW Sydney, viewed 10 August 2026.